

GDP and CO₂: the cost of economic growth

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1 Introduction

The economic growth of a country has always been closely related to increased energy consumption and environmental exploitation. However, recent growing investments in renewable energy, together with technological progress, could have loosened this relationship. For this reason, in this project we try to examine the link between economic activity, measured by Gross Domestic Product, and carbon dioxide emissions. Our purpose is to assess whether an association between GDP and CO₂ emissions exists and, if it does, if it persists over time, meaning that economic expansion continues to be accompanied by higher emission levels.

2 Dataset

We chose the dataset “CO₂ and Greenhouse Gas Emissions”, compiled by Hannah Ritchie, Pablo Rosado, and Max Roser and published by Our World in Data. This dataset contains annual information for a wide range of countries, spanning from 1750 to 2022. As the reader can imagine, data availability is not uniform across periods, in particular for earlier years and smaller countries. A schematic overview of the variables used in our analysis, together with the corresponding abbreviations adopted in the R script, is provided below:

- **data**: all the data in the table
- **mycountry_data**: all data relative to a selected country in the time period desired
- **year**: the year
- **country**: the country we take into account
- **gdp**: the Gross Domestic Product
- **co2_per_capita**: CO₂ emissions per capita

3 Data Cleaning

The first step we took was making the dataset clearer and more easily manageable by renaming the columns as we did in the previous paragraph. We also only took into consideration the time span going from 1980 to 2022 (to avoid unavailable data) and we only considered the CO₂ emissions per capita, in order to get a result which isn't totally dominated by the size of the country. We limited our analysis to the countries Italy, Brazil, Russia, India, China and South Africa. This choice is not random: Brazil, Russia, India, China and South Africa, known as BRICS, are developing economies that are growing in economic, as well as geopolitical, power at an astonishing rate. Italy was added to the analysis since us (the authors of this paper) are Italian and, moreover, Italy has been considered a well developed economy in the last decades.

4 Visual Analysis

In this section we will analyse the evolution of Gross Domestic Product and CO₂ emissions over time for the selected countries. Let the reader notice that we rescaled all the GDP data of a factor of 10^{12} , to use trillions of dollars as unit of measure.

The following graphs show the evolution of Gross Domestic Product over time for the selected six countries.

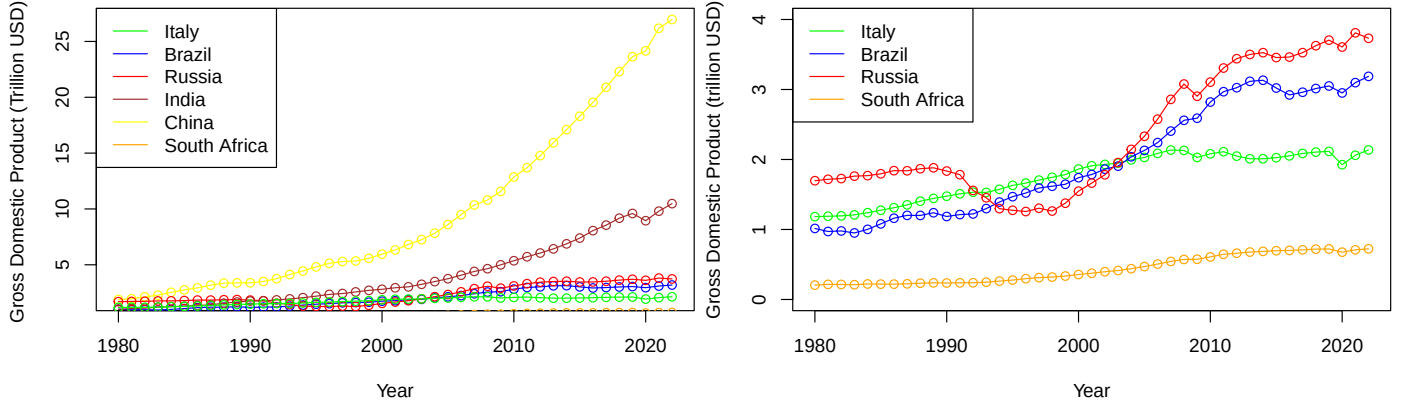


Figure 1: Left plot shows GDP evolution over time for Italy and BRICS, according to the legenda. Right plot shows a zoom of the same data for values of GDP below 4 trillion USD.

We can clearly notice that all countries have increased their GDP with respect to 1980, some keeping an increasing rate over the years (like Brazil, South Africa, India, and China). Another couple of important remarks have to be made: although all the economies we selected start relatively close, India and China have grown at a rate that far surpasses all the others. Another observation regards the small dip in Russia's GDP around 1990, which can be explained by the dissolution of the USSR in 1991; the same problem is to be seen in the following lines, where we will discuss CO₂ emissions per capita (since Russia has, as of 2022, about half the inhabitants USSR had).

Using the same colored notation as before, we take a look at the carbon dioxide emissions per capita for the selected six countries.

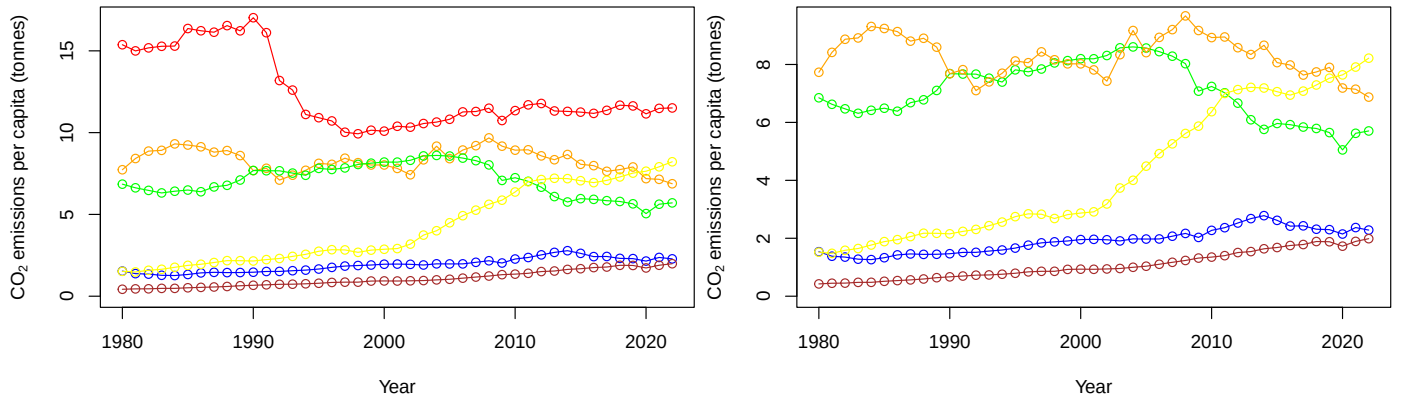


Figure 2: The graph on the left represents CO₂ emissions per capita over time, while the graph on the right shows a zoom of the same plot for values of emissions below 10 tonnes

We can see that the selected economies have different trends: while some countries' trends show a decrease in carbon dioxide emissions, for example Italy, Russia or South Africa; other countries like India or China are growing in their emissions numbers; but did the average carbon dioxide emission level actually change over the years? (See section 5). Now the next logical question is: is there a relation between the two? We try to answer that by looking at scatter plots showing the relationship between GDP and CO₂ emissions.

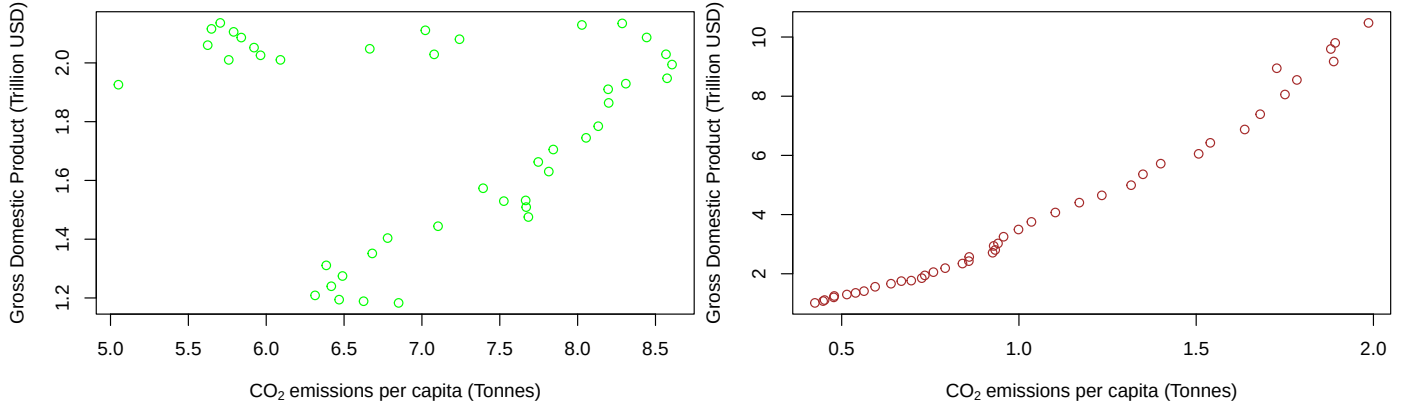


Figure 3: The plot on the left shows the correlation between CO₂ emissions and GDP for Italy (green), while on the right we have the same plot for India (brown)

We can see that for Italy, in green, there doesn't appear to be a clear relationship between the two quantities, while for India, in brown, the relationship seems to be clearly linear, i.e. the two are growing proportionally. We include only these two plots to avoid repetition, since the plots of China and Brazil show a similar pattern to the Indian one, and the South African and Russian plots do not show an evident relationship between the two (see Appendix A).

5 Hypothesis Testing

To completely answer the question asked in the section above (whether the mean CO₂ emission level has changed over time periods) we design the following hypothesis:

$$H_0 : \mu_{\text{early}} = \mu_{\text{late}} \quad \text{versus} \quad H_1 : \mu_{\text{early}} \neq \mu_{\text{late}}.$$

where μ is the average CO₂ emission level and we defined “late” as the period 2000-2022, while “early” refers to the period 1980-1999.

This two-sample T-test yields $t = -0.682$ and a p-value of $p = 0.496$; therefore we do not have enough statistically significant evidence to reject the null hypothesis at level $\alpha_0 = 0.05$, i.e. we cannot conclude that the average CO₂ emission level changed significantly in the two time periods (we include the whole R code output in Appendix C).

6 Regression

We proceed to model the relationship between GDP and CO₂ emissions using linear regression (although our research question concerns the existence of a relationship between economic growth and CO₂ emissions, we model GDP as the dependent variable). The preliminary visual inspection of the data using scatter plots reveals that the relationship between GDP and CO₂ emissions is highly nonlinear for some countries. Moreover, both variables span several orders of magnitude. To address these issues, since both variables are strictly positive, we can apply a logarithmic transformation to both GDP and CO₂ emissions, which allows the relationship to be interpreted in proportional terms. This practice is standard in economics, as economic output is commonly characterized by approximately exponential growth over time, and taking the logarithm helps to linearize, making a linear regression framework more appropriate. In the appendix, the reader can find the fitted values-residuals plots, both for the simple model and for the log-scaled model, and see that the variance becomes more stable with the transformation, confirming our choice (see Appendix B for the simple regression plots).

6.1 Simple Linear Regression

We begin by studying the direct relationship between GDP and CO₂ emissions using the following model:

$$\log(\text{GDP}_{it}) = \beta_0 + \beta_1 \log(\text{CO}_{2,it}) + e_{it},$$

where i indexes countries and t indexes years.

This regression yields the results showed in Appendix C. A brief summary is provided below:

The estimated coefficient on $\log(\text{CO}_2)$ is $\hat{\beta}_1 = 0.75$, with a p-value smaller than 2×10^{-16} . The model exhibits a high goodness of fit, with an R^2 of approximately 0.60, indicating that a large fraction of the variation in GDP is associated with variation in CO₂ emissions (see Appendix B).

6.2 Multiple Linear Regression

To assess whether this relationship has changed over time, we extend the baseline specification by including a time variable and an interaction term between $\log(\text{CO}_2)$ and year_c . This allows the elasticity of GDP with respect to emissions to vary over time. We do this by estimating the following model:

$$\log(\text{GDP}_{it}) = \beta_0 + \beta_1 \cdot \log(\text{CO}_{2,it}) + \beta_2 \text{year}_c + \beta_3 \cdot [\log(\text{CO}_{2,it}) \cdot \text{year}_c] + e_{it},$$

where $\text{year}_c = \text{year} - 1980$, so that $\text{year}_c = 0$ corresponds to the base year 1980.

As we did for the simple regression, we include the whole R code output in Appendix C, while we provide a brief explicative summary below:

The estimated coefficient on $\log(\text{CO}_2)$ is $\hat{\beta}_1 = 0.46$, which represents the elasticity in the base year 1980. It has very low p-value: 7.5×10^{-10} . The interaction coefficient is estimated as $\hat{\beta}_3 = 0.01$ and is highly statistically significant (p-value = 0.0006). Also this model has a high goodness of fit: $R^2 = 0.69$, slightly higher than the simple regression one.

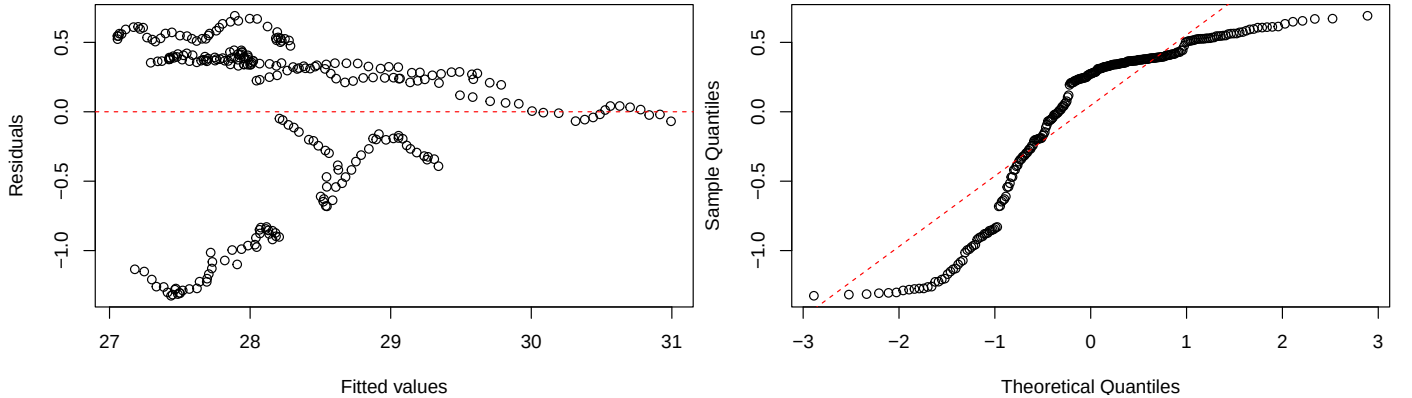


Figure 4: On the left we have the Fitted Values vs Residuals for the multiple regression, while on the right we have the Q-Q plot for the same regression model.

The residuals versus fitted values plot shows that residuals are generally centered around zero across the range of fitted values, although their dispersion varies and some structured patterns are visible. Residuals are not uniformly scattered, suggesting that variability changes with the level of the fitted values. The Q-Q plot indicates that residuals follow reasonably well the theoretical normal distribution in the central region, while noticeable deviations are visible in the lower and upper tails. This behaviour suggests that the residual distribution differs from normality at the extremes.

7 Discussion

We start by inspecting the simple regression. The corresponding hypothesis test is:

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_1 : \beta_1 \neq 0.$$

Given the extremely small p-value, we decisively reject the null hypothesis and conclude that there is strong statistical evidence of a positive association between GDP and CO₂ emissions at the aggregate level.

We now discuss the multiple regression. The relevant hypothesis for testing whether a GDP–CO₂ relationship exists over time is:

$$H_0 : \beta_3 = 0 \quad \text{versus} \quad H_1 : \beta_3 \neq 0.$$

The positive and significant estimate of β_3 leads us to discard the inference of a declining relationship. The multiple regression model achieves an R^2 of approximately 0.69 and a lower residual standard error than the simple regression, indicating an improvement in model fit.

8 Interpretation

The results of the two regression models both tend to imply a persistent association between Gross Domestic Product and Carbon Dioxide emissions. Following the simple regression model, we found that economic growth is closely linked to carbon-intensive activities. In fact, the estimated elasticity shows that 1% increase in CO₂ emission is associated, on average, with an approximately 0.75% increase in GDP (we remind the reader that in our analysis we considered the logarithmic transformation of both GDP and CO₂). The multiple regression model indicates that this relationship strengthens over time. As proof of that, we start with a 46% elasticity of GDP with respect to CO₂ emissions in the base year (1980) and it is increasing over time, as the estimated elasticity of the interaction is positive and statistically significant. We also have to consider that in our data observations for a given country might not be independent over time: both values of GDP and CO₂ emissions in one year are naturally related to those observed in previous years. For this reason, regression errors for a country may be serially correlated across years, violating the assumption of independence across observations. This doesn't invalidate the estimated coefficients, but it might reduce the precision of the estimated standard errors and affect statistical inference. Finally we, as authors, want to stress that this result is not to be interpreted as a causation between the two, only as a correlation (a statistically significant one). We leave a more geopolitical reading of these findings to the next dedicated section.

9 Conclusion

To end our analysis, we can say that we have statistically significant evidence that there is a positive correlation between economic growth and carbon dioxide emissions, although we don't have enough statistically significant evidence to conclude that this correlation is decreasing over time. On a more geopolitical note, we have to say that this data is not as conclusive as one could hope for many reasons: first of all the data we took into consideration doesn't account for technological innovation, differences in energy structure and in institutional quality; moreover this dataset doesn't mention the amount of green energy that each country produces. On that last note, we have to state that India and China, two of the most polluting countries in the world, are actually undertaking, successfully, a green transition: as of 2024 solar energy amounts for about 10% of their energetical source (according to Ember Energy Research). In conclusion we can say that, although statistically we found some important evidence supporting correlation, the analysis of the relationship between CO₂ and GDP remains a topic which is far too complicated to be addressed only by raw data, as the world is.

Appendix

In this section we include plots and R code which we used for our preliminary analysis.

A - GDP AND CARBON DIOXIDE EMISSIONS

The first plots we include are the scatter plots for China (yellow) and Brazil (blue). Below we find Russia (red) and South Africa (orange).

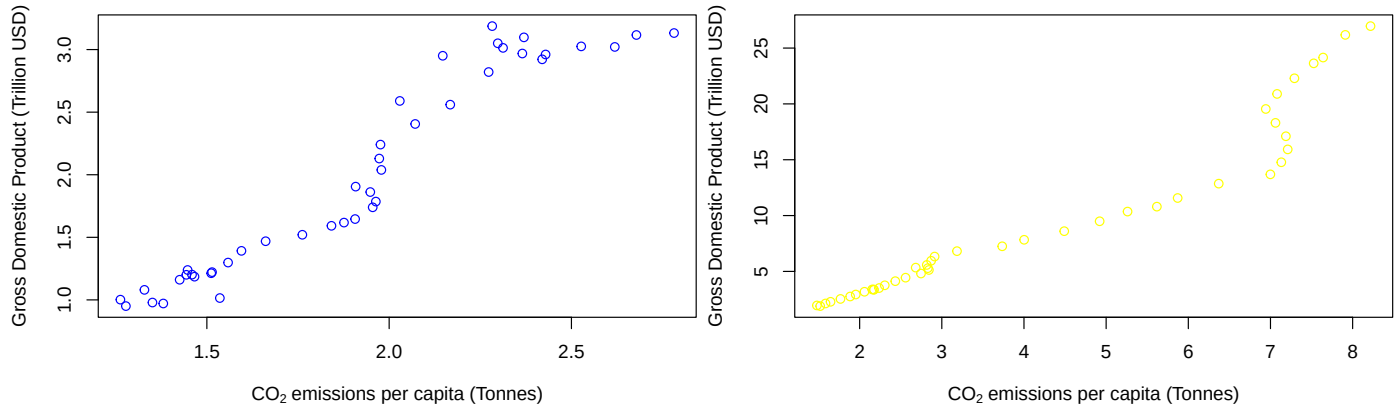


Figure 5: These plots show a behaviour similar to the one of India, both with a correlation coefficient well above 0.95

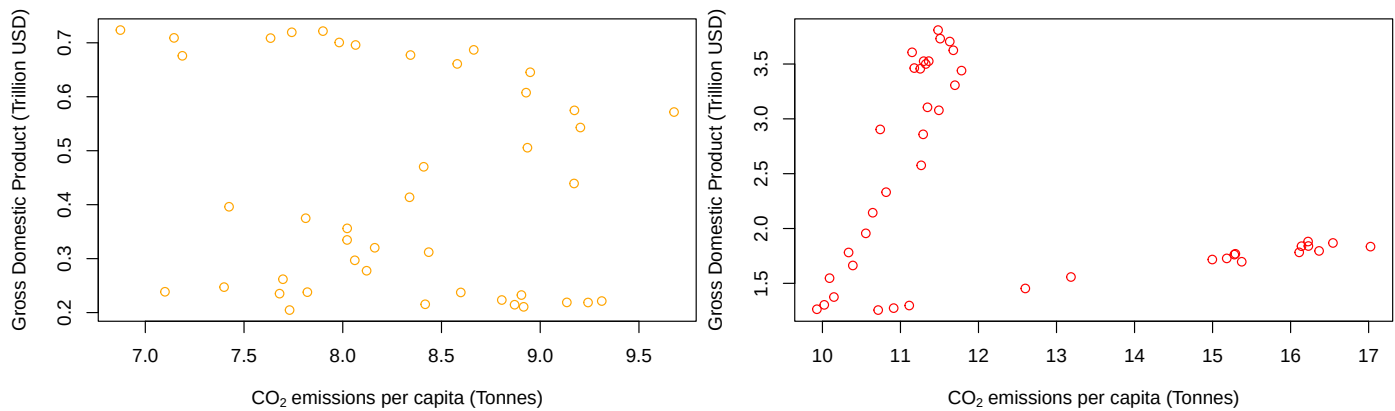


Figure 6: These plots show a behaviour similar to the Italian one, both with a correlation coefficient around -0.2

B - LOGARITHMIC TRANSFORMATION

Now we show why we needed to apply the logarithmic transformation to our regression model, to avoid repetitions we only include the graphs for the simple regression:

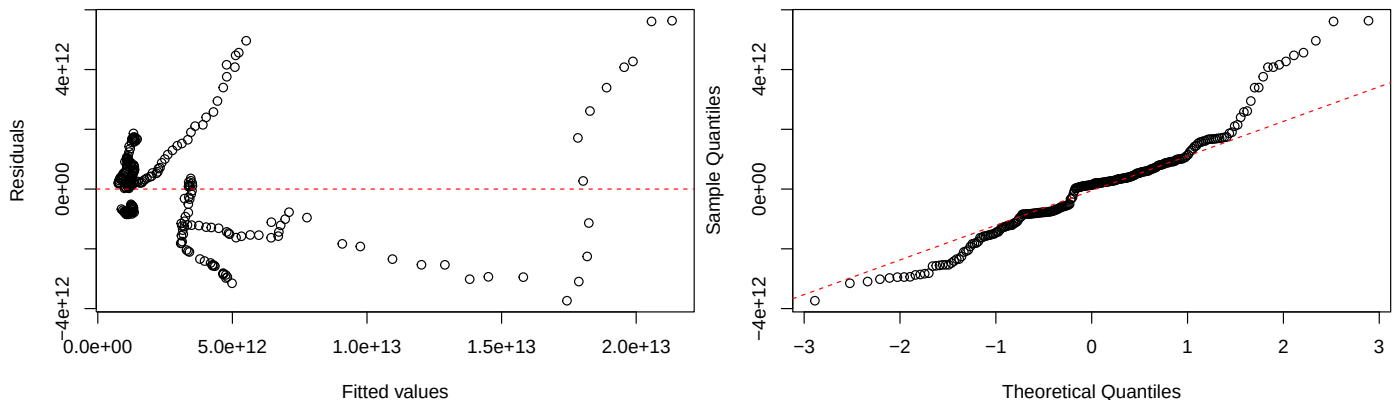


Figure 7: On the left we can see the Fitted values vs Residual graph for the simple regression, without the logarithmic transformation, on the right we can see the Q-Q plot of residuals for the same regression mode

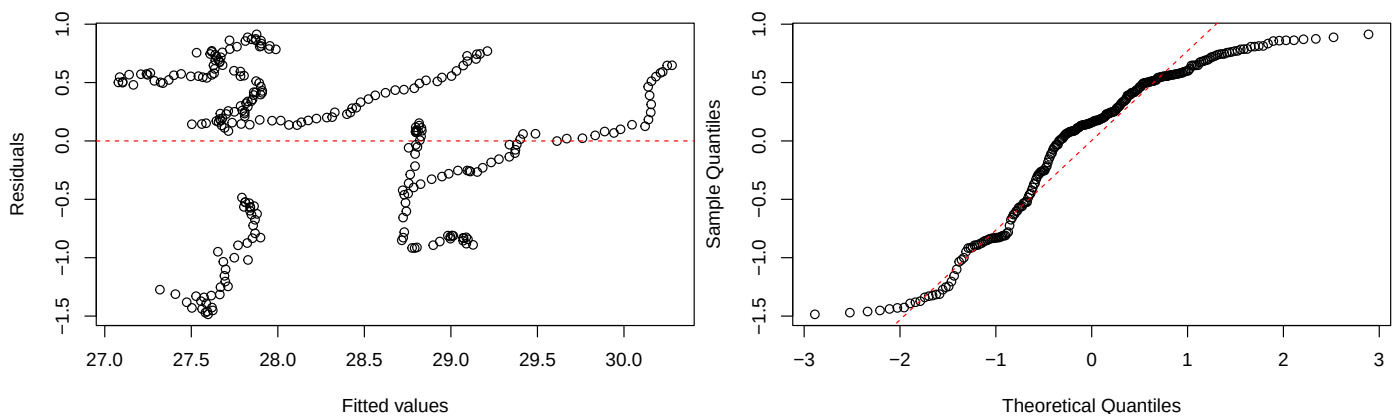


Figure 8: On the left we can see the Fitted values vs Residual graph for the simple regression, with the logarithmic transformation; on the right we can see the Q-Q plot of residuals for the same regression model

We can then notice that the non-logarithmic transformation gives more disperse values compared to the transformed one, thus justifying our choice.

C - R CODE OUTPUTS

We now include the whole R code output for the statistical test seen in section 5.

```
##
## Welch Two Sample t-test
##
## data: early_co2 and late_co2
## t = -0.68175, df = 213.16, p-value = 0.4961
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -1.4155713 0.6880137
## sample estimates:
## mean of x mean of y
## 5.640794 6.004573
```

We now provide the regression model summary for the simple regression, already briefly summarized in section 6.1:

```
##
## Call:
## lm(formula = log(gdp) ~ log(co2), data = data_countries)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.4851 -0.5141  0.1582  0.5214  0.9128
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  23.2384     0.2618   88.77  <2e-16 ***
## log(co2)      0.7514     0.0383   19.62  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6473 on 256 degrees of freedom
## Multiple R-squared:  0.6006, Adjusted R-squared:  0.599
## F-statistic: 385 on 1 and 256 DF, p-value: < 2.2e-16
```

Below the reader can see the same summary, for the multiple regression model:

```
##
## Call:
## lm(formula = log(gdp) ~ log(co2) + year_c + log(co2):year_c,
##     data = data_countries)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3255 -0.2972  0.2742  0.3883  0.6915
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   24.678211   0.470430  52.459  < 2e-16 ***
## log(co2)       0.460679   0.071995   6.399  7.5e-10 ***
## year_c        -0.039381   0.018085  -2.178  0.030358 *
## log(co2):year_c 0.009287   0.002675   3.472  0.000606 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5772 on 254 degrees of freedom
## Multiple R-squared:  0.685, Adjusted R-squared:  0.6813
## F-statistic: 184.1 on 3 and 254 DF, p-value: < 2.2e-16
```

D - BIBLIOGRAPHY

- <https://ourworldindata.org/co2-and-greenhouse-gas-emissions>
- <https://github.com/owid/co2-data>
- <https://edition.cnn.com/2025/11/07/climate/solar-wind-renewables-transition-global-pakistan-hungary-chile>
- <https://en.wikipedia.org/wiki/BRICS>